

Miles Franck

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SUMMARY

Cybersecurity-focused software engineer with hands-on SOC analyst experience (BCR Cyber) and a technical foundation in Python, database systems, and secure software development. Practical exposure to SIEM tools, vulnerability analysis, and threat modeling, with CompTIA Security+ in progress. Seeking to join a collaborative engineering team where I can apply a security-aware mindset to real-world systems.

CORE COMPETENCIES

Threat Modeling • Vulnerability Assessment • Security Analysis • Secure SDLC • Cryptography Fundamentals
• Data Sanitization • Agile / Scrum Collaboration • Technical Documentation

SOFTWARE & SECURITY SKILLS

Security Tools (Familiar): Shodan, Nmap, Splunk, Wireshark

Languages & Databases: Python, PostgreSQL, MySQL, Rust

Systems & Tooling: Git, WSL

EXPERIENCE

Summer Fellow — Carnegie Mellon University, Heinz College Summer Fellowship *2025*

- Completed coursework in network defense, database systems, and information security, including a threat intelligence report on a specific threat actor and a critical infrastructure vulnerability assessment
- Engaged with industry and government security professionals as part of coursework, including representatives from Dragos and the FBI
- Partnered with an external organization on a team SIEM evaluation project, gathering stakeholder requirements and researching tool capabilities (log ingestion, alerting, usability, cloud support)
- Delivered a live presentation of findings to an audience that included CMU staff and outside industry guests

SOC Operations Analyst II — BCR Cyber *Dec 2025*

- Participated in a live-fire cyber range exercise, identifying and tracing an active attacker's presence within a simulated enterprise environment

SOC Operations Analyst I — BCR Cyber *Nov 2025*

- Performed security operations tasks building on foundational SOC analyst training, including log analysis, alert triage, and incident documentation workflows.

Teaching Assistant — Allegheny College *2022 – 2025*

- Provided one-on-one technical support and debugging help to students during office hours, translating complex concepts into clear, actionable guidance.
- Reviewed student code and projects for correctness and adherence to secure coding and best-practice standards.
- Communicated across technical and non-technical audiences to align grading criteria and project expectations with faculty.

PROJECTS

BanjoDB *2025*

- Built a dual-language architecture separating a secure Rust core from a Python interface to emulate enterprise-style backend isolation.
- Mitigated credential theft risk by implementing Argon2 hashing with unique per-user salts.

Deepseeking-Vulns *2025*

- Developed an AI-assisted vulnerability assessment workflow using Shodan, Ollama, and DeepSeek to analyze exposed network services and support penetration-testing preparation and risk analysis.

Automated Vulnerability Analysis Tool *2024*

- Built a Python tool that queries the Shodan API to enumerate internet-facing assets and cross-references results against CVE/CPE vulnerability databases.

Rust TCP Chat Server *2026*

- Built a multi-client TCP chat server in Rust using the Tokio asynchronous runtime, supporting concurrent client connections and real-time message broadcasting.
- Implemented a publish-subscribe messaging system using Tokio broadcast channels to efficiently distribute messages to all connected clients.
- Designed a room management system with unique join codes, user authentication, connection limits, and participant tracking to control access to each session.

CERTIFICATIONS

- CompTIA Security+ — CompTIA (Expected October 2026)
- SOC Operations Analyst II — BCR Cyber (Dec 2025)
- SOC Operations Analyst I — BCR Cyber (Nov 2025)
- Network Defenses 95-884 — Carnegie Mellon University, Heinz College (2025)